In [2]:
```
import numpy as np
```

In [3]:
```
np.__version__
```

Out[3]:
```
'1.26.4'
```

In [4]:
```
my_list = [0,1,2,3,4,5,6,7,8]
my_list
```

Out[4]:
```
[0, 1, 2, 3, 4, 5, 6, 7, 8]
```

In [5]:
```
type(my_list)
```

Out[5]:
```
list
```

In [6]:
```
arr = np.array(my_list) # convert list to array
arr
```

Out[6]:
```
array([0, 1, 2, 3, 4, 5, 6, 7, 8])
```

In [7]:
```
type(arr)
```

Out[7]:
```
numpy.ndarray
```

In [8]:
```
np.arange(10)
```

Out[8]:
```
array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

In [9]:
```
np.arange(10,15)
```

Out[9]:
```
array([10, 11, 12, 13, 14])
```

In [10]:
```
np.arange(10,20,5)
```

Out[10]:
```
array([10, 15])
```

In [11]:
```
np.zeros(5) #parameter tuning
```

Out[11]:
```
array([0., 0., 0., 0., 0.])
```

In [12]:
```
np.zeros(5, dtype=int) #hyperparameter tuning
```

Out[12]:
```
array([0, 0, 0, 0, 0])
```

In [13]:
```
np.zeros([2,2], dtype=int) #2d array
```

Out[13]:
```
array([[0, 0],
       [0, 0]])
```

In [14]:
```
np.zeros([5,10], dtype=int) # nd array
```

Out[14]:
```
array([[0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0]])
```

In [15]:
```
np.ones(3)
```

Out[15]:
```
array([1., 1., 1.])
```

In [16]:
```
np.ones([3,5], dtype = int)
```

Out[16]:
```
array([[1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1]])
```

In [17]:
```
np.arange(10,20)
```

Out[17]:
```
array([10, 11, 12, 13, 14, 15, 16, 17, 18, 19])
```

In [18]:
```
np.arange(20,10)
```

Out[18]:
```
array([], dtype=int32)
```

In [19]:
```
np.arange(-20,10)
```

Out[19]:
```
array([-20, -19, -18, -17, -16, -15, -14, -13, -12, -11, -10,  -9,  -8,
        -7,  -6,  -5,  -4,  -3,  -2,  -1,   0,   1,   2,   3,   4,   5,
         6,   7,   8,   9])
```

In [20]:
```
random.rand(3,2)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[20], line 1
----> 1 random.rand(3,2)

NameError: name 'random' is not defined
```

In [23]:
```
np.random.rand(2)
```

Out[23]:
```
array([0.96320024, 0.5289118 ])
```

In [25]:
```
np.random.rand(5)
```

Out[25]:
```
array([0.35737395, 0.23822721, 0.66140198, 0.0860246 , 0.19874466])
```

In [27]:
```
np.random.rand(3,2)
```

Out[27]:
```
array([[0.8957197 , 0.84578692],
       [0.95370265, 0.47060447],
       [0.09126194, 0.24179451]])
```

In [29]:
```
np.random.randint(3)
```

Out[29]: 1

In [31]:
```
np.random.randint(5,10)
```

Out[31]: 9

In [33]:
```
np.random.randint(5,10,3)
```

Out[33]: array([5, 9, 8])

In [35]:
```
ran = np.random.randint(5,10,3)
print(ran)
```

```
[7 9 6]
```

In [37]:
```
otp = np.random.randint(5,10,4)
print(otp)
```

```
[8 8 7 5]
```

In [39]:
```
arr1 = np.random.randint(10,40,(10,10))
print(arr1)
```

```
[[27 33 23 36 39 22 35 12 30 20]
 [26 16 23 38 35 28 27 26 14 23]
 [31 20 17 10 35 39 26 26 29 21]
 [34 30 12 26 36 32 11 31 10 20]
 [34 14 13 10 19 22 33 36 14 39]
 [32 29 10 17 36 16 14 38 26 25]
 [30 22 23 17 26 39 20 31 34 26]
 [13 38 13 12 23 27 30 27 36 14]
 [21 30 16 15 33 36 33 15 11 24]
 [33 39 17 33 18 20 16 13 18 32]]
```

In [41]:
```
arr2 = np.random.randint(5,10,(10,10)) #10*10 matrix along with value between 5 to
print(arr2)
```

```
[[9 5 7 6 8 6 5 7 9 6]
 [7 6 5 9 7 5 5 5 7 7]
 [5 5 6 5 7 7 9 9 6 7]
 [7 5 8 6 8 9 7 5 9 9]
 [9 8 5 5 6 5 6 6 5 5]
 [5 7 5 7 9 8 7 8 9 6]
 [8 5 7 9 5 7 6 7 6 5]
 [5 6 7 9 7 9 5 5 7 6]
 [8 5 9 7 7 7 9 5 5 7]
 [8 9 8 5 9 7 8 7 6 5]]
```

In [43]:
```
arr3 = np.random.randint(1,5,(3,3))
print(arr3)
```

```
[[2 4 1]
 [2 4 3]
 [1 4 4]]
```

indexing, slicing, array

In [46]:
```
arr3[:]
```

Out[46]:
```
array([[2, 4, 1],
       [2, 4, 3],
       [1, 4, 4]])
```

In [48]:
```
len(arr3)
```

Out[48]:
```
3
```

In [50]:
```
arr3.shape
```

Out[50]:
```
(3, 3)
```

In [52]:
```
arr3[0]
```

Out[52]:
```
array([2, 4, 1])
```

In [54]:
```
arr3[1]
```

Out[54]:
```
array([2, 4, 3])
```

In [56]:
```
arr3[:,1]
```

Out[56]:
```
array([4, 4, 4])
```

In [58]:
```
arr3[:,2]
```

Out[58]:
```
array([1, 3, 4])
```

In [60]:
```
arr1
```

Out[60]:
```
array([[27, 33, 23, 36, 39, 22, 35, 12, 30, 20],
       [26, 16, 23, 38, 35, 28, 27, 26, 14, 23],
       [31, 20, 17, 10, 35, 39, 26, 26, 29, 21],
       [34, 30, 12, 26, 36, 32, 11, 31, 10, 20],
       [34, 14, 13, 10, 19, 22, 33, 36, 14, 39],
       [32, 29, 10, 17, 36, 16, 14, 38, 26, 25],
       [30, 22, 23, 17, 26, 39, 20, 31, 34, 26],
       [13, 38, 13, 12, 23, 27, 30, 27, 36, 14],
       [21, 30, 16, 15, 33, 36, 33, 15, 11, 24],
       [33, 39, 17, 33, 18, 20, 16, 13, 18, 32]])
```

In [62]:
```
arr1[-8]
```

Out[62]:
```
array([31, 20, 17, 10, 35, 39, 26, 26, 29, 21])
```

In [64]:
```
arr1[:,-8]
```

Out[64]:
```
array([23, 23, 17, 12, 13, 10, 23, 13, 16, 17])
```

In [66]:
```
arr1[3:4, 4:6] #3:4 belongs to row & 4:6 belongs to columns
```

Out[66]:
```
array([[36, 32]])
```

In [79]:
```
arr3[1,2]
```

Out[79]:
```
3
```

In [81]:
```
arr3[1:2]
```

Out[81]:
```
array([[2, 4, 3]])
```